AHC VISUAL INTELLIGENCE HACKATHON · 5 SEPT 2026

Real-time video anomaly detection

Can a **small** vision-language model watch live video
and report what matters — fast enough, and without crying wolf?

59.7

out of 100 · was 49.9 one hour before the end

29.8 / 35

our strongest tier — 4-minute videos localised well

6.1×

faster than real time · 1.65 s per 10 s window

ARCHITECTURE

What runs, and what is new at each stage.

Ingest
28 videos · 5 s – 10 min

NEW
Sampled **BY TIMESTAMP**, never frame index — sources run 1.9 to 30 fps, so indices mis-time every event

WINDOWS
20 s wide, 10 s hop. Short clips get one window covering the whole clip

Appearance model
Qwen3-VL-8B · 16 frames

PROMPTING
Grouped fine-grained questions — traffic, hazard, behaviour — not one flat "what's wrong here"

OUTPUT
Constrained JSON: class, span, confidence, plain-language reason

INSTRUMENTED
Every call timed at source — p50 / p95 / max, not a config number

Duration model
SigLIP2 frozen · 8 frames

NEW
Frozen encoder → 8 vectors → **BIDIRECTIONAL GRU** reads them **IN SEQUENCE**

WHY
One frame of a person standing is normal. Eight frames of them not moving is loitering. A per-frame classifier cannot see this

Fuse & shape
CPU · no GPU

NEW · UNION
Never replace. Either model may add a class the other missed

NEW · SPAN GEOMETRY
Long videos: merge same-class events, pad ±120 s. **WORTH +6.1 MARKS ALONE**

PER-DIFFICULTY
Short clips top-2 · 4-min spans untouched · policy differs by tier, measured not assumed

Output
59.7 / 100

VALIDATED
Schema checked before every upload — ids, null timestamps at tier 1, spans inside duration

THREE STATES
no row = not run · **FAILED** = the run broke · empty = model says normal. Never blurred

technique we introduced · standard practice, applied carefully

Two models, one merge, zero retraining at the end. The last +9.8 marks came from the fuse-and-shape stage — **CPU only, no GPU, no changed class.**

WHY TWO MODELS

Over half the events came from the encoder, not the VLM.

Source of the 38 events we shipped	Events
SigLIP2 only — the VLM never saw these	**17**
Both models agreed	10
Qwen3-VL only — SigLIP never saw these	7
Derived by merging overlapping spans	4

Only **10 of 38** events had both models agreeing. The other 28 came from exactly one of them.
That is the case for running two — not a story, a count.

MODELS TESTED

Five VLMs, identical test. Two surprises.

Model	Size	Precision	Recall	Verdict
Qwen3-VL-8B-Instruct	**8B**	**0.62**	**0.40**	**Shipped**
Qwen3-VL-4B-Instruct	4B	0.77	0.50	Ties the 8B
Cosmos-Reason2-8B	8B	0.57	0.20	Half the recall of a 4B
Qwen3-VL-2B-Instruct	2B	0.30	0.15	Too small
Qwen2.5-VL-7B-Instruct	7B	0.00	0.00	Silent on 20 of 24
SigLIP2 + trained GRU head	**frozen**	**added +2.4 marks**		**Shipped**

Cosmos-Reason2-8B is purpose-built for traffic anomaly reasoning — and got half the recall of a general 4B here.
SigLIP was marked "rejected" in our own ledger, on theory. It became one of the two things that actually moved our score.

THE BREAKTHROUGH · FINAL HOUR

+9.8 marks. We changed no model.

Eight prediction sets — three model sizes, five window configs, a fine-tune, targeted prompts — all scored **exactly the same** on the long videos.

That invariance was the clue. If changing the model changes nothing, the failure isn't in the model.

Our predicted spans were **1–8 seconds** inside videos of **327–602 seconds**. A 5-second span can never overlap a 100-second event by half — **geometrically impossible, however right the class is.**

49.9 → 56.0

widening spans to 112–138 s.
No new inference. No changed class.

Our classes had been right all along.

→ 59.7

adding SigLIP2, which predicts

loitering

— a class we score 0/6 on and

no VLM proposed all day.

The evidence was in our own notes from that morning: the sample data held a **125-second** event. We wrote down that these events are long — then spent hours swapping models instead of spans.

HOW WE WORKED

Three AI agents, one repository.

Agent	Role	Owned
Coordinator	Research, strategy, every GPU run, every commit	inference lane, Modal driver
Builder	Data pipeline, synthetic video generation, training, scoring	data lane, tests
Comms	Live demo, slides, process record	demo, docs

- **Ownership by directory.** No agent edits another's files. We learned it the hard way — two agents built the same four modules before the rule existed.
- **A written ledger, not chat.** Every technique marked done / running / rejected — **with the reason**. An agent could crash, restart and resume from files. It happened twice.
- **They corrected each other.** One caught a scorer using the wrong matching rule. One caught a run that exited "successfully" having done 6 of 34 videos.

WHAT WE LEARNED

Ten failures in one day. None of them crashed.

- **ffmpeg's seek silently returns more than you ask for.** A 468 s video rendered as 241 s — every timestamp after it was wrong.
- **The fix for one bug caused another.** We stopped one bad video killing a 34-video run; it then wrote 10 failures to disk as confident, plausible "nothing here".
- **A model reported 100% validation accuracy.** It had 7 validation samples, because an upload had died unnoticed.
- **We deleted 125 videos we had already rendered** — a third of our synthetic training data — rather than train on labels we could not verify.
- **We retracted our own headline finding** before it shipped. The retraction was load-bearing: keeping it meant shipping the weaker model.

> In a seven-hour build, **every failure that mattered was silent.**
> So we validated intermediate artifacts — not outputs.